EXAMPLES:

- Evaluate $\int_1^3 3x^2 dx$

Solution:

$$\int_1^3 3x^2 dx = [x^3]_1^3 = (3)^3 - (1)^3 = 27 - 1 = 26$$

- Evaluate $\int_2^3 (x^2 + x) dx$

Solution:

$$\int_2^3 (x^2 + x) dx = \left[\frac{x^3}{3} + \frac{x^2}{2}\right]_2^3 = \left(\frac{3^3}{3} + \frac{3^2}{2}\right) - \left(\frac{2^3}{3} + \frac{2^2}{2}\right) = \frac{53}{6}$$

EXAMPLE:

Find the area enclosed by the curves $y = x^2$ and $y^2 = 8x$.

SOLUTION:

$$y^2 = x^4 = 8x$$
$$x^4 - 8x = 0$$
$$x(x^3 - 8) = 0$$
$$x = 0 \;\; or \; x^3 - 8 = 0$$
$$x^3 = 8$$
$$x = 2$$

Thus, the curves intersect at $x = 0$ and $x = 2$.

## Assignment

- Use the trapezium rule, with ordinates at $x = 1, 2, 3, 4, 5$ to calculate, correct to two decimal places, an approximate value of

$$\int_1^5 \sqrt{(2x + 8x^{-2})^3}\, dx$$